Logging In SSH Troubleshooting

SSH Troubleshooting

Connection Refused or Timeout

Make sure you are using the correct login command:

```
ssh USERNAME@orcd-login.mit.edu
```

If the connection hangs or times out, you may be on a network that blocks SSH, or your IP address has been blocked (see below). Try a different network or contact orcd-help@mit.edu.

Permission Denied

Check your username: Make sure you are using your MIT Kerberos username.

Check your SSH key permissions: Run these commands on your laptop:

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/id_rsa
```

Key not uploaded: If you have not yet set up SSH keys, see the SSH Key Setup page.

> **Note**
>
> Even with SSH keys set up, you will still be prompted for your MIT Kerberos password and Duo. This is expected.

Host Key Warning

If you see a message like `WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED` or `This host key is known by the following other names`, run:

```
ssh-keygen -R orcd-login.mit.edu
```

Then reconnect and type `yes` when prompted. This is sometimes happens after system maintenance.

> **Warning**
>
> If this happens repeatedly without any maintenance, contact orcd-help@mit.edu.

Too Many Authentication Failures

If you see:

```
Received disconnect from ... : Too many authentication failures
```

Add the following to your `~/.ssh/config`:

```
Host orcd-login.mit.edu
    IdentityFile ~/.ssh/id_rsa
    IdentitiesOnly yes
```

> **Note**
>
> This setting only applies to connections to `orcd-login.mit.edu` and will not affect SSH connections to any other servers.

Forgot SSH Key Passphrase

There is no way to recover a forgotten passphrase. Generate a new SSH key pair and upload it to Engaging following the SSH Key Setup instructions.

Dropped or Slow Connections

Add the following to your `~/.ssh/config` to keep sessions alive:

```
Host orcd-login.mit.edu
    ServerAliveInterval 60
    ServerAliveCountMax 10
```

If your connection is slow to establish — particularly when connecting from off-campus — also add `GSSAPIAuthentication no`:

```
Host orcd-login.mit.edu
    ServerAliveInterval 60
    ServerAliveCountMax 10
    GSSAPIAuthentication no
```

By default, SSH attempts Kerberos (GSSAPI) authentication before trying your password or SSH key. This requires a handshake with MIT's Kerberos infrastructure, which can add several seconds to your connection time if you are on a slow or off-campus network. Disabling it tells SSH to skip straight to password and key authentication. This has no effect on users who do not use Kerberos tickets to authenticate.

Other factors that can affect connection stability

- **Wi-Fi instability** — a flaky wireless connection will drop SSH sessions regardless of keepalive settings. Try a wired connection if you experience frequent drops.
- **MIT VPN** — being connected to the MIT VPN can add latency depending on your network path.
- **Server-side idle timeout** — the server may enforce its own idle timeout. Client-side keepalives help but are not a guarantee against disconnection.
- **Large data transfers** — data transfers to or from your local machine can saturate your network connection and make your SSH session appear slow or unresponsive.

Cannot Connect to a Compute Node

Compute nodes are not directly accessible from outside the cluster, and you must have a running job on the node to connect to it. SSH to the login node first, then connect to the compute node:

```
ssh USERNAME@orcd-login.mit.edu
ssh nodeXXXX
```

To connect directly from your laptop in one step using ProxyJump, see Setting up your Config File in the VS Code recipe for an example config.

IP Address Blocked

If you have had repeated failed login attempts, our systems may have temporarily blocked your IP address. This will look like a connection timeout or `Connection refused` rather than an authentication error — your connection will not reach the server at all.

If you suspect your IP address has been blocked, stop attempting to connect and wait 10 minutes. The block will lift automatically if no further failed attempts are made during that time.

> **Warning**
>
> If you have VS Code open with a Remote SSH session, it may attempt to reconnect automatically in the background, which will reset the 10-minute timer and keep the block in place. Close VS Code before waiting for the block to clear.

If the block does not clear after 10 minutes, open a ticket with orcd-help@mit.edu and include your IP address. You can find it with:

macOS/Linux

```
curl ifconfig.me
```

Windows

```
curl ifconfig.me
```

If that does not work, try:

```
(Invoke-WebRequest -Uri "ifconfig.me").Content
```

Account Locked

10 failed Duo attempts will lock your account. This lock is managed by MIT IS&T, not ORCD, and will lift automatically after 90 minutes provided no further failed login attempts are made during that time.

> **Warning**
>
> If you have VS Code open with a Remote SSH session, it may attempt to reconnect automatically in the background, resetting the 90-minute timer and keeping your account locked. Close VS Code before waiting for the lock to clear.

To prevent this from happening again:

- Log into ORCD OnDemand — this satisfies Duo authentication and removes the Duo requirement for SSH for a few hours.
- Use SSH Control Channels to reduce how often you need to authenticate with Duo.
- If you use VS Code, adjust the VS Code Remote SSH settings to limit reconnection attempts.

If your account does not unlock after 90 minutes, contact orcd-help@mit.edu.

Still Having Trouble?

If you have worked through this guide and are still unable to connect, run the following command and include the output in your ticket to orcd-help@mit.edu:

```
ssh -vvv USERNAME@orcd-login.mit.edu
```

The `-vvv` flag enables verbose output and gives the support team detailed information about where the connection is failing.

June 10, 2026